QUBO Formulation: Benchmark Job Selection Problem

Benchmark Job Selection Problem

Problem Statement. Given a set of N candidate benchmark jobs, each characterized by a coverage value v_i, a compute time t_i, and pairwise redundancy penalties p_{ij}, select a subset of jobs to maximize the total benchmark coverage value while penalizing redundant coverage, subject to a total compute-time budget T.

Decision Variables. For each candidate job $i \in \{0, \ldots, N-1\}$, introduce a binary variable $x_i \in \{0, 1\}$, where $x_i = 1$ if job i is selected, and $x_i = 0$ otherwise.

QUBO Derivation. Step 1. The original objective is a maximization problem. To cast it as a QUBO minimization, we negate the objective function:

$$\min_{x \in \{0,1\}^N} \left(-\sum_{i=0}^{N-1} v_i x_i + \sum_{0 \le i < j \le N-1} p_{ij} x_i x_j \right). \tag{1}$$

Step 2. The compute-time budget is expressed as a linear inequality constraint:

$$\sum_{i=0}^{N-1} t_i x_i \le T. \tag{2}$$

Step 3. We enforce the constraint via a quadratic penalty term with weight $\lambda > 0$:

$$P(x) = \lambda \left(\sum_{i=0}^{N-1} t_i x_i - T \right)^2. \tag{3}$$

Expanding the square and applying the idempotent property $x_i^2 = x_i$ for binary variables yields:

$$P(x) = \lambda \left(\sum_{i=0}^{N-1} t_i^2 x_i^2 + \sum_{i \ne j} t_i t_j x_i x_j - 2T \sum_{i=0}^{N-1} t_i x_i + T^2 \right) \tag{4}$$

$$= \lambda \left(\sum_{i=0}^{N-1} (t_i^2 - 2T t_i) x_i + 2 \sum_{0 \le i < j \le N-1} t_i t_j x_i x_j + T^2 \right). \tag{5}$$

The general penalty contribution to the Hamiltonian is therefore:

$$P(x) = \sum_{i=0}^{N-1} \lambda(t_i^2 - 2Tt_i)x_i + 2\lambda \sum_{0 \le i < j \le N-1} t_i t_j x_i x_j + \lambda T^2. \tag{6}$$

Step 4. Combining the negated objective and the penalty term gives the full QUBO Hamiltonian $H(x)$:

$$H(x) = \sum_{i=0}^{N-1} \left(-v_i + \lambda(t_i^2 - 2Tt_i)\right) x_i + \sum_{0 \le i < j \le N-1} \left(p_{ij} + 2\lambda t_i t_j\right) x_i x_j + \lambda T^2. \tag{7}$$

Step 5. Reading off the upper-triangular QUBO matrix entries $Q_{i,j}$ and the constant offset c from $H(x) = \sum_{i \le j} Q_{i,j} x_i x_j + c$, we obtain the closed-form expressions:

$$Q_{i,i} = -v_i + \lambda(t_i^2 - 2Tt_i), \quad i \in \{0, \ldots, N-1\}, \tag{8}$$

$$Q_{i,j} = p_{ij} + 2\lambda t_i t_j, \quad 0 \le i < j \le N-1, \tag{9}$$

$$Q_{i,j} = 0, \quad i > j, \tag{10}$$

$$c = \lambda T^2. \tag{11}$$

Penalty Weight Justification. The penalty weight λ must be chosen large enough so that the cost of violating the budget constraint strictly exceeds any possible gain in the benchmark coverage objective. Let $V_{\max} = \max_i v_i$. A sufficient condition is $\lambda > V_{\max}$ (or more conservatively $\lambda > \sum_i v_i$). This ensures that any assignment violating $\sum_i t_i x_i \le T$ incurs a penalty larger than the maximum achievable objective value, thereby forcing the global minimum of $H(x)$ to lie within the feasible region.

Correctness Argument. Minimizing $H(x)$ over $\{0,1\}^N$ recovers the optimal solution to the original problem. For any feasible assignment satisfying $\sum_i t_i x_i \le T$, the penalty term $P(x)$ evaluates to zero, and $H(x)$ reduces exactly to the negated original objective. Thus, minimizing $H(x)$ over feasible points is equivalent to maximizing the original objective. For any infeasible assignment, $P(x)$ is strictly positive. By choosing λ according to the justification above, the energy of every infeasible point exceeds the energy of the optimal feasible point. Consequently, the global minimum of $H(x)$ must correspond to a feasible assignment that maximizes the original objective function.

Illustrative Example. Consider an instance with $N = 3$ jobs. We instantiate the parameters as follows: coverage values $v_0 = 10, v_1 = 15, v_2 = 12$; redundancy penalties $p_{01} = 2, p_{02} = 3, p_{12} = 2$; compute times $t_0 = 3, t_1 = 4, t_2 = 5$; budget $T = 8$; and penalty weight $\lambda = 100$. Substituting these values into the general closed-form expressions

yields the concrete QUBO coefficients:

$$Q_{0,0} = -10 + 100(3^2 - 2 \cdot 8 \cdot 3) = -3910, \tag{12}$$
$$Q_{1,1} = -15 + 100(4^2 - 2 \cdot 8 \cdot 4) = -4815, \tag{13}$$
$$Q_{2,2} = -12 + 100(5^2 - 2 \cdot 8 \cdot 5) = -5512, \tag{14}$$
$$Q_{0,1} = 2 + 2 \cdot 100 \cdot 3 \cdot 4 = 2402, \tag{15}$$
$$Q_{0,2} = 3 + 2 \cdot 100 \cdot 3 \cdot 5 = 3003, \tag{16}$$
$$Q_{1,2} = 2 + 2 \cdot 100 \cdot 4 \cdot 5 = 4002, \tag{17}$$
$$c = 100 \cdot 8^2 = 6400. \tag{18}$$

The resulting Hamiltonian is:

$$H(x) = -3910x_0 - 4815x_1 - 5512x_2 + 2402x_0x_1 + 3003x_0x_2 + 4002x_1x_2 + 6400. \tag{19}$$

Evaluating $H(x)$ over all feasible assignments (those with $\sum_i t_i x_i \leq 8$) gives:

$$H(0,0,0) = 6400, \quad H(1,0,0) = 2490, \quad H(0,1,0) = 1585, \quad H(0,0,1) = 888, \tag{20}$$
$$H(1,1,0) = -923, \quad H(1,0,1) = -19. \tag{21}$$

The assignment $x^* = (1,1,0)$ yields the minimum energy $H(x^*) = -923$, corresponding to selecting jobs 0 and 1. This matches the optimal benchmark coverage value of $v_0 + v_1 - p_{01} = 10 + 15 - 2 = 23$, confirming the formulation's correctness on this instance.